

Kinematics 1D and 2D

* There need to be frame of reference from where observer will analyse the state of rest and motion. observer with coordinate axis

* we attach a coordinate system to observer that is called frame of reference.

* There are two type of frame of reference

(a) Inertial frame o.r (when observer $\rightarrow$ Rest, const velocity)

(b) Non inertial frame o.r (when observer $\rightarrow$ acce. (const, non-uniform))

* The change in coordinate of the system describe the type m.

* Distance - Total path covered irrespective of direction.

* Displacement - Shortest path from Initial to final
Direction is very important

* Distance = $R\theta$ $\rightarrow$ for circle)

* Displacement = $2R \sin \theta/2$

Dimensi, onless $\theta = \frac{\text{Arc length}}{\text{Radius}}$ rad

* Instantaneous -

* Speed = distance/time

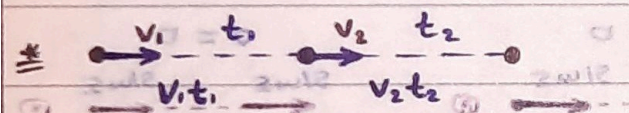
velocity = displac/time

Rate of change of distance w.r.t time $\frac{m}{s}$ $\frac{L}{t}$

Rate of change of displac w.r.t time $v = \frac{dx}{dt}$

* Speed = 0, +ve

* velocity = +ve, -ve, 0



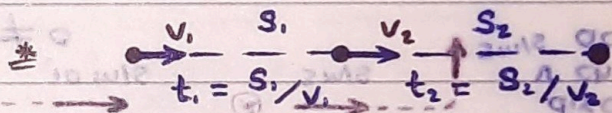
$$\text{Avg. Speed} = \frac{\text{total distance}}{\text{total time}} = \frac{v_1 t_1 + v_2 t_2}{t_1 + t_2}$$

* If $t_1 = t_2$ $V_{av} = \frac{v_1 + v_2}{2}$

uniform speed

Equal distance in equal time

* 100 km/h (time independent)



$$\text{Avg. Speed} = \frac{\text{Total distance}}{\text{total time}} = \frac{S_1 + S_2}{t_1 + t_2} = \frac{S_1 + S_2}{S_1/v_1 + S_2/v_2}$$

* If $S_1 = S_2$ $V_{avg} = \frac{2v_1 v_2}{v_1 + v_2}$

non uniform speed

when distance travelled changes w.t

* $\text{Speed} = 4t^2 \text{ m/s}$, $4x^2$

* Some definitions for velocity (distance $\rightarrow$ displacement)

* Average velocity

$$a_v = \frac{\text{Total disp}}{\text{total time}}$$

* Average speed

$$a_v = \frac{\text{Total distance}}{\text{total time}}$$

* Instantaneous velocity

velocity at particular instant

$$v = \frac{dx}{dt} \bigg|_{t=t_1}$$

* Instantaneous speed

the mag of velocity at that time

$$\text{Inst speed} = |v| = \left| \frac{dx}{dt} \right|_{t=t_1}$$

* $A \rightarrow 3 \text{ m/s}$ $\neq$ Speed $A = 3 \text{ m/s}$ $\neq V_A = +3 \text{ m/s}$ $V_A \neq V_B$
 $3 \text{ m/s} \leftarrow B$ $\neq$ Speed $B = 3 \text{ m/s}$ $\neq V_B = -3 \text{ m/s}$

* Can particle have a constant speed and variable velocity.

Ans. Speed = constant $\neq$ Velocity $\neq$ const (becoz of direction)

* What does speedometer of car measure? $\rightarrow$ Inst Speed

$\odot$ velocity at any time = $\pm$ (Instantaneous speed)

* $\xrightarrow{x_1} \xrightarrow{x_2}$ Avg Speed = $x_1 + x_2 / \text{total time}$

Avg Velocity = $\frac{x_1 - x_2}{T \cdot \text{time}}$ $\neq$ when particle in 1D, if it changes direction.

Avg speed $\neq$ |Avg velocity|

$\odot x = -t^2$ $\therefore t_1 = 1$ Speed $t = 1$ $|V| = 2 \text{ m/s}$
 $V = \frac{dx}{dt} = -2t$ $V = -2 \text{ m/s}$ $\therefore t_2 = 2$ $|V| = 4 \text{ m/s}$
 $\therefore$ If $t = 2$ $V = -4 \text{ m/s}$

* Acceleration - Rate of change of velocity w.r.t time. m/s^2

(*) $\xrightarrow{3 \text{ m/s}} \xrightarrow{3 \text{ m/s}}$ $a = 0$ $\odot \xrightarrow{3 \text{ m/s}} \xrightarrow{10 \text{ m/s}}$ $a \neq 0$ $\odot \xrightarrow{3 \text{ m/s}}$ $\downarrow 3 \text{ m/s}$ dir change $V \text{ change}$ $\text{acc} \neq 0$

* Uniform Acce - when equal change in velocity in equal time. 3 m/s^2

* Non uniform Acce - when there is unequal change in velocity in equal time. $\text{ex} - a = 3t^2, -4t^2$

* Instantaneous Acce -

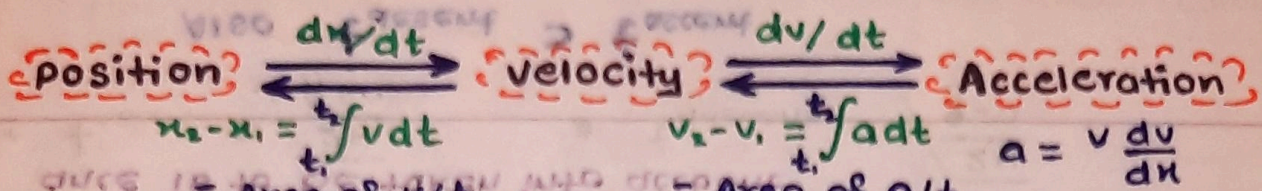
* Average Accele

acc. at particular Instant of time. $a = \frac{dv}{dt} \bigg|_{t=t_1}$

$a_{av} = \frac{\text{change in velocity}}{\text{total time}}$

- when a particle moves with non-uniform speed then acce. of the particle be non-zero.
- The direction of avg. acceleration vector is the direction of the change in velocity vector as $\vec{a} = \Delta \vec{v} / \Delta t$
- Acce which opposes the motion of body is called retardation.
- Acceleration has same direction of force.
- Retardation $\Rightarrow$ velocity and acce are opposite in direction.
- Acceleration $\Rightarrow$ velocity and acce are same in direction.
- sign of velocity (+ve or -ve) represents the direction of motion but sign of acce indicates the direction of accele.
- If velocity and acce both are having same sign, then magnitude of velocity (i.e Speed) is increasing and if both have opposite signs then magnitude of velocity (i.e Speed) is decreasing.

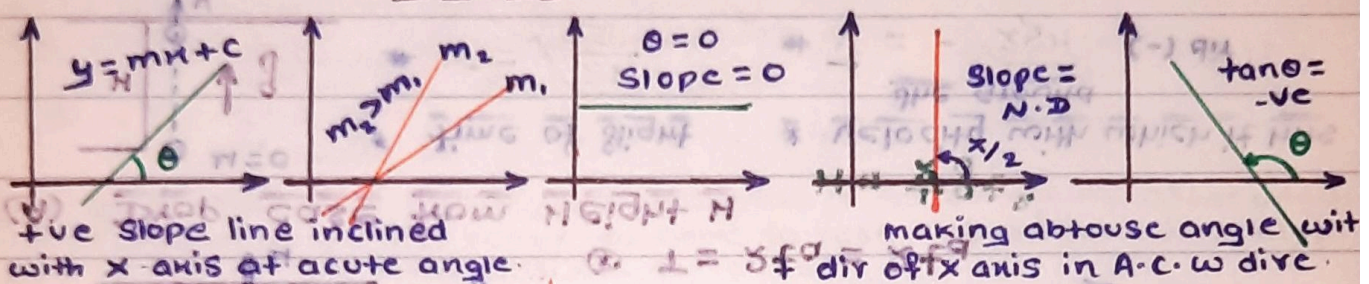
KINGDOMS ID AND SD



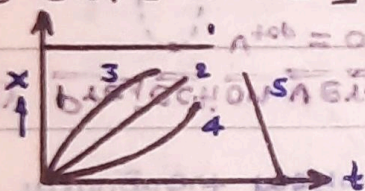
- $v = u + at$ $a = \text{acceleration} = \text{constant}$
 - $S = ut + \frac{1}{2}at^2$ $S = \text{displacement}$ $S = vt - \frac{1}{2}at^2$
 - $v^2 - u^2 = 2aS$ $S_{nth} = \text{Displacement in the } n^{\text{th}} \text{ sec.}$
 - $S_{nth} = u + \frac{1}{2}a(2n-1)$ $\textcircled{*}$ If a is variable $a = \frac{dv}{dt}$ $a = v \frac{dv}{dx}$
- $\hookrightarrow$ means what displac he made in that particular second

$$S = v_{av} t = \left(\frac{u+v}{2} \right) t$$

Graphs



$\textcircled{*}$ Slope of this graph represents instantaneous velocity.



- $v = 0$ (Rest)
- $v = +ve$ (uniform) constant
- $v = +ve$ (Speed $\downarrow$) NON-uniform motion
- $v = +ve$ (Speed $\uparrow$) NON-uniform motion
- $v = -ve$ (Speed = const) uniform

$\neq$ Area of $x-t$ graph = $\int x dt$ = no physical significance.

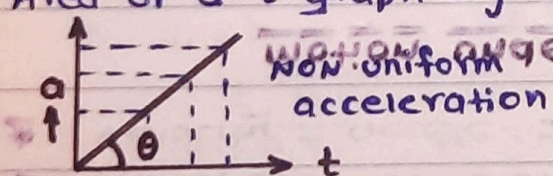
$\neq$ Area of this graph = $x_2 - x_1$ = change in position.

$\textcircled{*}$ All above cases for velocity-time graph also. (same)

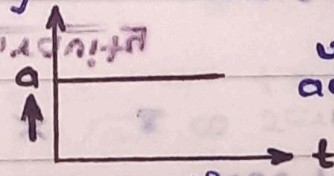
$\neq$ Area of $v-t$ graph = $\int v dt$ = displac = change in position

Acceleration-time graph

Area of $a-t$ graph = $\int a dt = \int dv = v_2 - v_1$ = change in velocity



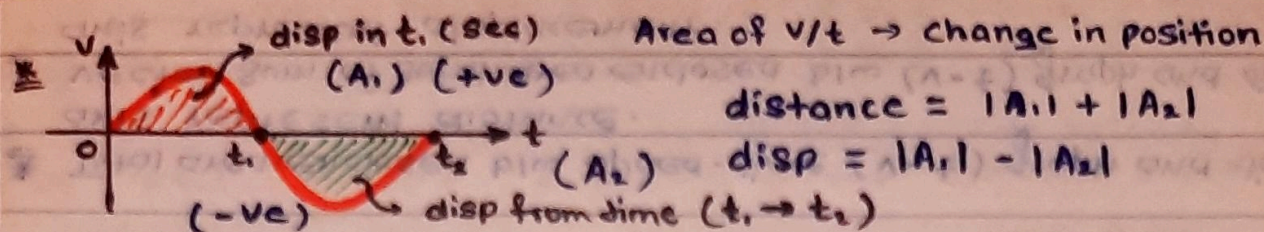
i.e uniformly $\uparrow$ acceleration



$\neq$ i.e uniform or const acceleration.

$\neq$ Total area enclosed b/w speed-time ($v-t$) graph and time axis represent distance.

$\neq$ vector sum of total area enclosed b/w ($v-t$) graph and time axis represent displacement.



* -ve speed, $\text{dist} = -ve \rightarrow$ Not possible

* $\text{velocity} = \text{accele} = \infty$ * $\infty \text{ Jerk} =$ Not possible

Motion under Gravity

* Any Qty pointing upwards (+)

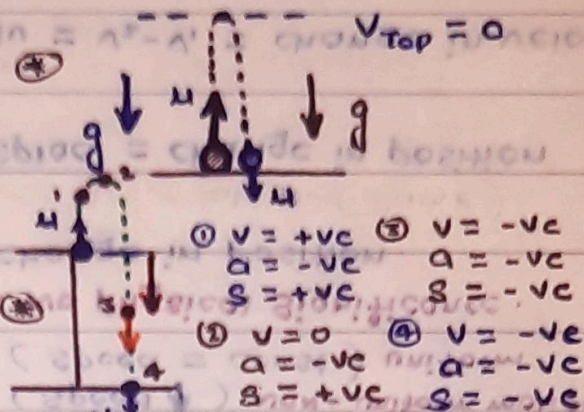
* any Qty pointing downward (-)

* we will consider point of projection as origin.

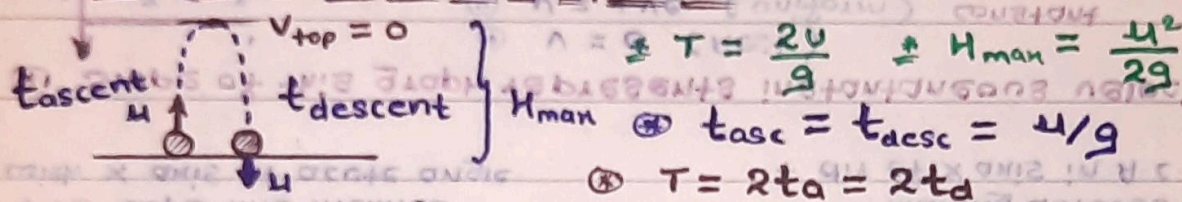
* neglect air friction and variation in Gravity

* Projection speed and Landing speed is same at same level

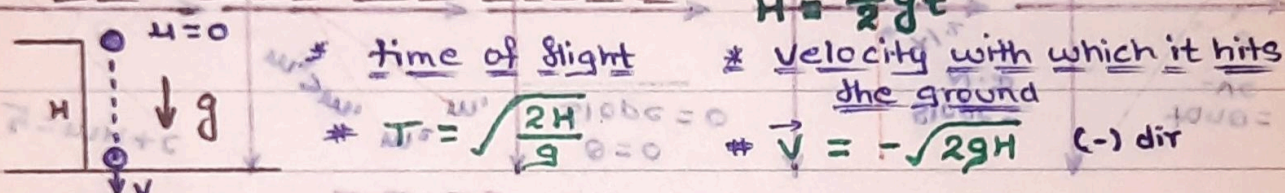
* Throughout motion $\text{acc} = -g \hat{j}$



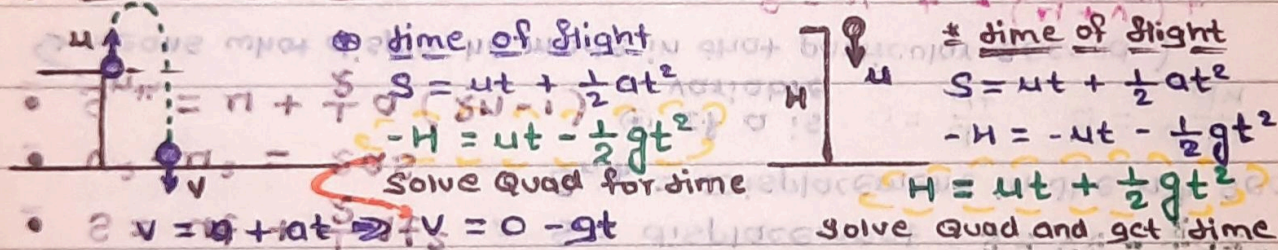
(A) Projection vertically upward



(A) Drop case from Height H



(A) Projection vertically upward from top of a tower



* A body is thrown vertically upwards, if constant Air resistance is to be taken into account.

$t_{\text{ascent}} = \sqrt{\frac{2H}{g+a}}$ $t_{\text{descent}} = \sqrt{\frac{2H}{g-a}}$
 Also $t_{\text{descent}} > t_{\text{ascent}}$

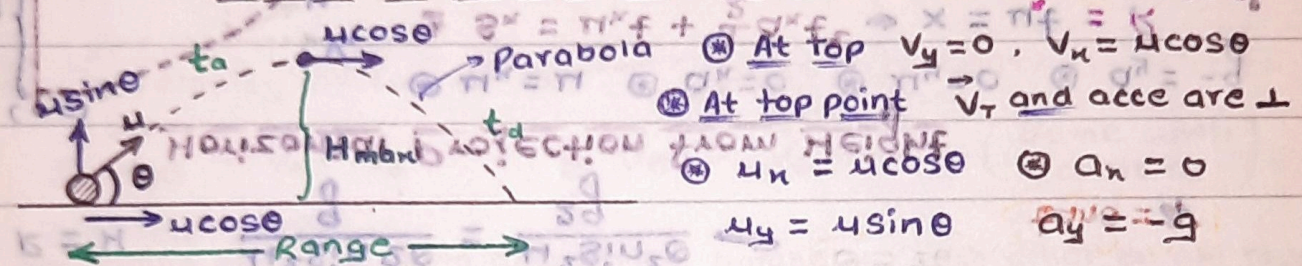
Kinematics in 2D

Ex If particle has initial velocity $2\hat{i} + 3\hat{j}$ and acceleration is $\hat{i} + 2\hat{j} \text{ ms}^{-2}$. Find velocity and position at $t = 1 \text{ s}$.

Ans $\vec{u} = 2\hat{i} + 3\hat{j}$ and $\vec{a} = \hat{i} + 2\hat{j}$
 $\therefore u_x = 2 \quad \therefore u_y = 3 \quad \therefore a_x = 1 \quad \therefore a_y = 2$

$\therefore \vec{v} = v_x\hat{i} + v_y\hat{j} \quad \therefore \vec{s} = s_x\hat{i} + s_y\hat{j}$
 $= (u_x + a_x t)\hat{i} + (u_y + a_y t)\hat{j} \quad = (u_x t + \frac{1}{2} a_x t^2)\hat{i} + (u_y t + \frac{1}{2} a_y t^2)\hat{j}$
 $= (2 + 1 \times 1)\hat{i} + (3 + 2 \times 1)\hat{j} \quad = (2 \times 1 + \frac{1}{2} \times 1 \times 1^2)\hat{i} + (3 \times 1 + \frac{1}{2} \times 2 \times 1^2)\hat{j}$
 $= 3\hat{i} + 5\hat{j} \quad = \frac{5}{2}\hat{i} + 4\hat{j}$

Ground to Ground Projection



Coordinates of Particle at time t

$y = u \sin \theta t - \frac{1}{2} g t^2 \quad \therefore y = \frac{u \sin \theta x}{u \cos \theta} - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$
 $t = \frac{x}{u \cos \theta} \quad y = x \tan \theta - \frac{g x^2}{2 u^2 \cos^2 \theta}$

At Any time $x = u \cos \theta t$ eqn of trajectory $y = x \tan \theta (1 - \frac{x}{R})$
 $y = u \sin \theta t - \frac{1}{2} g t^2$

Velocity of particle $\vec{v} = v_x\hat{i} + v_y\hat{j}$
 at time $t \quad = (u_x + a_x t)\hat{i} + (v_y + a_y t)\hat{j}$
 $\vec{v} = (u \cos \theta)\hat{i} + (u \sin \theta - g t)\hat{j}$

Change in velocity and momentum $\vec{p} = m\vec{v}$ of particle

$\vec{p} = m \cdot \text{velocity}$
 $\vec{u}_i = u \cos \theta \hat{i} + u \sin \theta \hat{j}$
 $\vec{v}_t = (u \cos \theta)\hat{i} + (u \sin \theta - g t)\hat{j}$
 $\Delta \vec{v} = \vec{v}_t - \vec{v}_i$
 $\Delta \vec{p} = m(\vec{v}_t - \vec{v}_i) = m(-g t)\hat{j}$

Time of Flight (T), max height (H), and Range (R)

$T = \frac{2 u \sin \theta}{g} \quad H_{\text{max}} = \frac{u^2 \sin^2 \theta}{2g} = \frac{u_y^2}{2g}$
 $R = \frac{u^2 \sin 2\theta}{g} = \frac{2 u_x u_y}{g}$

Comparison of two projectiles of equal range

① when projected at complementary angle



$$R_1 = R_2 \quad H_1 = \frac{u^2 \sin^2 \theta}{2g} \quad H_2 = \frac{u^2 \cos^2 \theta}{2g}$$

$$\frac{H_1}{H_2} = \tan^2 \theta \quad H_1 + H_2 = \frac{u^2}{2g} = \text{max Height in vertical proj}$$

$$H_2 - H_1 = \frac{u^2 \cos 2\theta}{2g} \quad R = 4\sqrt{H_1 H_2}$$

$$T_1 = \frac{2u \sin \theta}{g} \quad T_2 = \frac{2u \cos \theta}{g} \quad \frac{T_1}{T_2} = \tan \theta$$

$$T_1 \cdot T_2 = \left(\frac{u^2 \sin 2\theta}{g} \right) \times 2 \quad T_1 \cdot T_2 \propto R$$

Kinetic Energy of a Projectile

$$KE_i = \frac{1}{2} m u^2 \quad KE_{\text{top}} = \frac{1}{2} m u^2 \cos^2 \theta$$

$$KE_{\text{top}} = KE_i \cos^2 \theta$$

If $H_A = H_B$ then $(u_y)_A = (u_y)_B$ and $T_A = T_B \left(1 - \frac{u}{u} \right)$

Condition for maximum range when $\theta = 45^\circ$

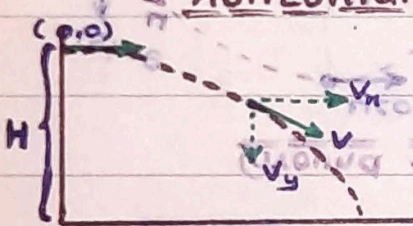
$$R = \frac{u^2 \sin 2\theta}{g} \quad \text{Range} = \frac{u^2}{g}$$

$$\sin 2\theta = 1 \quad \theta = 45^\circ \quad H = \frac{u^2 \sin^2 45^\circ}{2g} = \frac{u^2}{4g} \quad \sqrt{H} = \frac{R_{\text{max}}}{4}$$

Angle for range and height is same

$$R = H \quad \frac{u^2 \sin 2\theta}{g} = \frac{u^2 \sin^2 \theta}{2g} \Rightarrow \theta = 16^\circ \quad \tan \theta = 2$$

Horizontal Projection from Height



$$u_x = u \quad a_x = 0 \quad u_y = 0 \quad a_y = -g$$

$$s_x = u_x t + \frac{1}{2} a_x t^2 \Rightarrow x = ut = R$$

$$y = -\frac{1}{2} g \frac{x^2}{u^2} \Rightarrow y = -\frac{1}{2} g \frac{x^2}{u^2} \quad \text{eqn of trajectory}$$

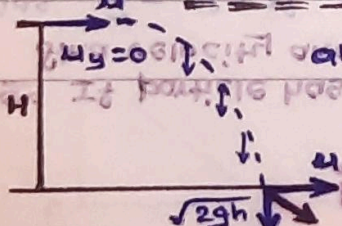
Velocity at a general point

$$\vec{v} = v_x \hat{i} + v_y \hat{j} = (u + a_x t) \hat{i} + (u_y + a_y t) \hat{j}$$

$$\vec{v} = (u) \hat{i} - gt \hat{j}$$

$$T = \sqrt{\frac{2H}{g}} \quad R = u \sqrt{\frac{2H}{g}} \quad \text{use } s_y = -H$$

Velocity after falling a height h



along x: ($a_x = 0$)

$$V_y^2 - u_y^2 = 2a_y s_y \quad V_{\text{net}} = \sqrt{u^2 + 2gh}$$

$$V_y^2 = -2g(-h)$$

Incline Projectile

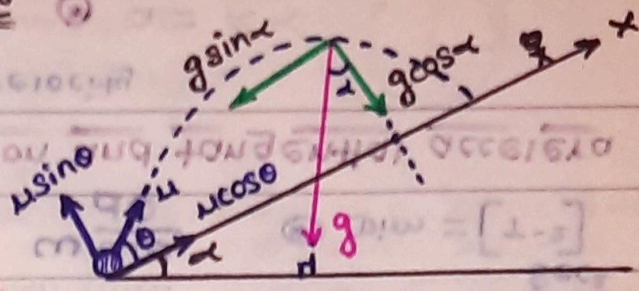
$\odot u_x = u \cos \theta$ $\odot u_y = u \sin \theta$

$\odot a_x = -g \sin \alpha$ $\odot a_y = g \cos \alpha$

$\ast T = \frac{2u \sin \theta}{g \cos \alpha} = \frac{2u_y}{a_y}$

$\ast H_{\text{max}} = \frac{2u^2 \sin^2 \theta}{g \cos \alpha} = \frac{u_y^2}{2a_y}$

$\ast \text{Range} = \frac{u \cos(\theta + \alpha) 2u \sin \theta}{g \cos \alpha}$



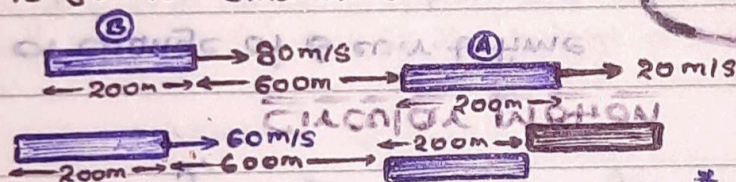
Relative Velocity

$\vec{V}_{AB}$ = velocity of w.r.t B $\longrightarrow$ observer (Rest)

$\ast \vec{V}_{AB} = \vec{V}_A - \vec{V}_B$ $\ast \vec{V}_{BA} = \vec{V}_B - \vec{V}_A$ (Same with accelera)

Q Two trains of length 200m are moving parallel to each other on rail roads such that A is ahead of B by 600m. If velocities of A = 20m/s and B is 80m/s. Find time in which B overtake A.

Ans



Let A is observer

$\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$
 $= 80\hat{i} - 20\hat{i} = 60\hat{i}$

$\ast \text{Speed} = 60 \text{ m/s}$ $\ast D = 600 + 200 + 200 = 1000 \text{ m}$

$\ast S = \frac{D}{S} = \frac{1000}{60} = \frac{50}{3} \text{ sec}$

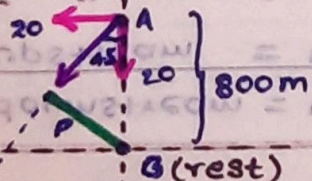
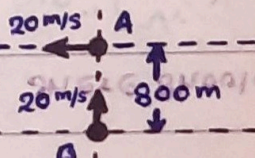
$\ast$ If $A_{\text{rel}AB} = 0$ and $A_{\text{rel}BA} = 0$ then their velocities w.r.t each other will be same.

$S = \frac{D}{T} \Rightarrow S_{\text{rel}} = \frac{D_{\text{rel}}}{T}$

Concept of closest distance of approach

$\ast$ Find closest distance of approach of A as seen by B

$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$
 $= -20\hat{i} - 20\hat{j}$



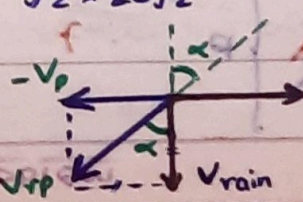
$\ast P = 800 \sin 45 = 800/\sqrt{2}$

$\ast$ Time taken to reach at closest point

$t = \frac{D}{S} = \frac{800}{\sqrt{2} \times 20/\sqrt{2}} = 20 \text{ sec}$

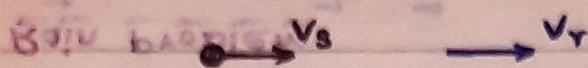
Rain Problem

$\vec{V}_{\text{rain person}} = \vec{V}_r - \vec{V}_p$
 $= \vec{V}_r + (-\vec{V}_p)$



$\ast \tan \alpha = \frac{V_p}{V_r}$

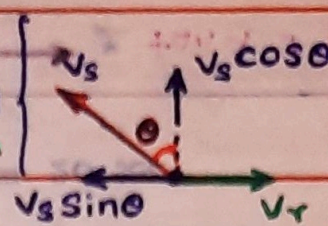
River Boat - Problem



downstream $= V_r = V_r + V_s$

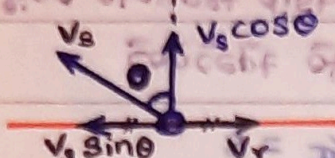
upstream $= V_r = V_s - V_r$

t_{min}
 $\cos \theta = \max$
 $\theta = 0^\circ$



time to cross river $t = \frac{d}{V_{app}} = \frac{d}{V_s \cos \theta}$

For shortest Path

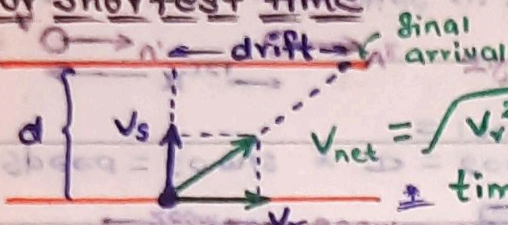


To reach directly opp point there should be no drifting in x.

$V_s \sin \theta = V_r \Rightarrow \theta = \sin^{-1}(V_r/V_s)$

time to reach $= d / V_s \cos \theta$

For shortest time



drift

$V_{net} = \sqrt{V_r^2 + V_s^2}$

time $= \frac{d}{V_s} = \frac{d}{V_s}$

* River will drift the body

$S_n = u_n t + \frac{1}{2} a_n t^2$

* $x = \frac{V_r d}{V_s}$

Circular Motion

Rate of change of θ w.r.t time

* $\omega = \frac{\theta_2 - \theta_1}{t_2 - t_1}$ * $\omega = \frac{d\theta}{dt}$

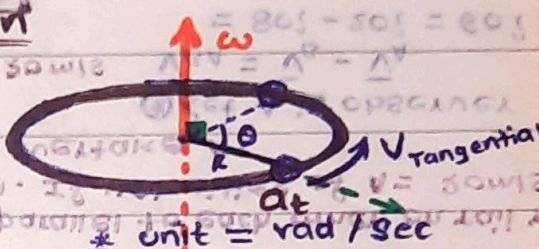
* Instantaneous $\omega = \frac{d\theta}{dt} \Big|_{t=t_1}$

* $\omega_{av} = \frac{\text{Total Angle Rotated}}{\text{Total time}}$

* Frequency (ν) $= \frac{1}{T}$

* $\omega = \frac{\text{Total Angle}}{\text{Time}} = \frac{2\pi}{T} = 2\pi \nu$

* $\vec{v} = \vec{\omega} \times \vec{R}$ * $v = R\omega \sin \theta$ * $v = R\omega$



* unit = rad/sec

* Dim = $[T^{-1}]$

→ Axial vector → Same as θ

ex $\rightarrow 360 \text{ rpm}$

$\nu = \text{rps} = \frac{\text{rpm}}{60} = \frac{360}{60}$

(θ is angle b/w R and ω)

Angular Acceleration (α) Rate of change of ω w.r.t time

$\alpha_{av} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\text{Total change in } \omega}{\text{Total time}}$ * unit = $\frac{\text{rad}}{\text{Sec}^2}$

Inst $\alpha = \frac{d\omega}{dt} = \frac{d\omega}{dt} \cdot \frac{d\theta}{d\theta} = \omega \frac{d\omega}{d\theta}$ * Dim = $[T^{-2}]$

Relation b/w angular acceleration and tangential accelera.

$a_t \Rightarrow$ Rate of change of Tangential velocity

* $a_t = \frac{dv}{dt}$ * $a = \alpha \times R$ * $a = R\alpha$

CH - Properties of Solid

Types of materials

"Perfectly elastic" "Inelastic" "Perfectly Inelastic"

③ which is more elastic? (a) Rubber (b) Steel

Easy to deform (x) ↙ ↘ Regaining is fast

what happens when external force act on body?

→ F_{ext} is applied → deformation of body

→ To regain its original shape body apply internal force

→ At eq^m $|F_i| = |F_{ext}|$

Stress = Force developed per unit Area inside body

Strain = change in confi / original configuration

Important Terms

Deforming force

At eq^m

Restoring force

F_{ext}

$$|F_{def}| = |F_{resto}|$$

Material → Inside force

Strain leads to Stress

Stress and its types

Longitudinal Stress

Shear Stress

Torsional Stress

force is $\perp$ to Area (Tangential Stress)

(Twisting)

Compressive

Tensile

Shape change

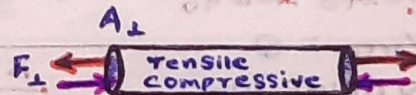
Normal Stress (Longitudinal Stress)

③ F is $\perp$ to Area

$$\text{Normal Stress} = \frac{F_{\perp}}{A_{\perp}} \left(\frac{N}{m^2} \right)$$

$$\text{dim} = [ML^{-1}T^{-2}]$$

Shear Stress



massless



Stress = 0

when Force applied is tangential to body → Shape changes

③ Case-① Normal Stress

Case-② Tangential Stress



$$\text{Comp. Stress} = \frac{F_{\perp}}{A_{\perp}}$$



$$\text{Shear Stress} = \frac{F_{\parallel}}{A_{\parallel}}$$

Volumetric Stress when force applied over a body changes Vol

$$\text{Volumetric Stress} = \frac{F_{\perp \text{ to Area}}}{\text{Area}} = \text{Pressure}$$

- ① Comp. Stress = $\frac{F_1}{A} = \frac{F \cos \theta}{A}$
- ② Shear Stress = $\frac{F \sin \theta}{A}$
- ③ massless Smooth Shear Stress = 0
- ④ Body will slip on surface

⑤ In real world no body is perfectly Rigid, so if we apply force, will there be deformation? Yes

Ans.

Strain (unitless and dimensionless)

Longitudinal Strain = $\frac{\Delta L}{L_0}$

Volumetric Strain = $\frac{\Delta V}{V}$

tan $\theta = \frac{\Delta x}{\Delta h}$ for small θ

$\theta = \frac{\Delta x}{\Delta h}$

Shear Strain = $\theta = \frac{\Delta x}{\Delta h}$

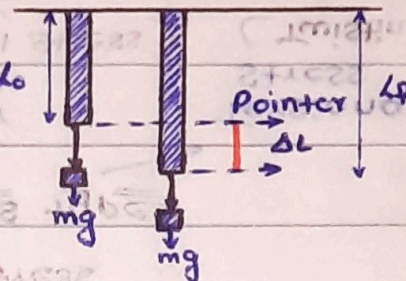
Comp (-) expan (+ve)

Hooke's Experiment

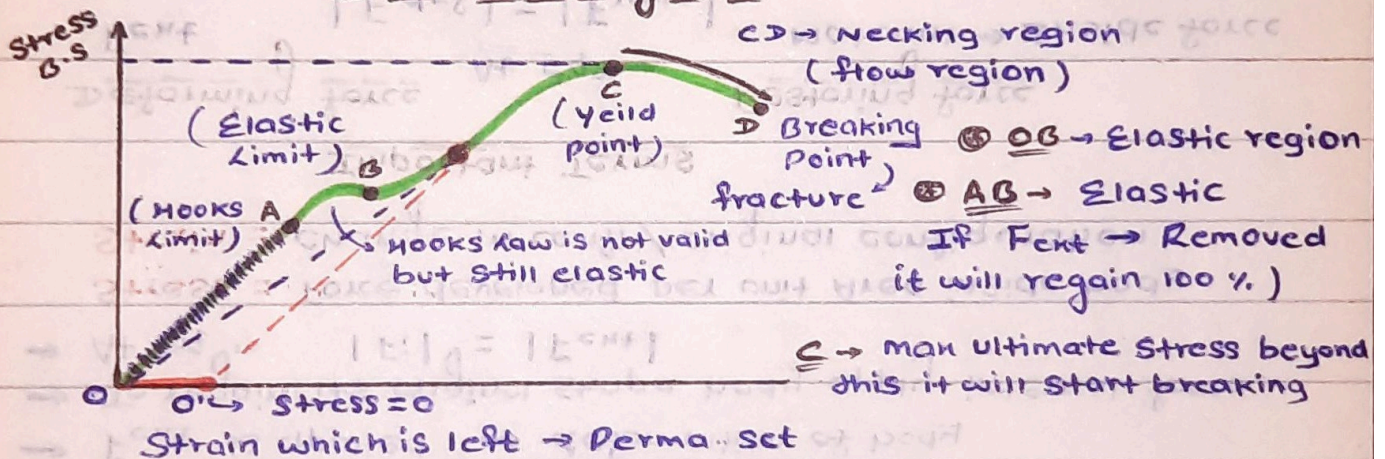
Relation b/w Stress and Strain develop

Stress = $\frac{F}{A} = \frac{mg}{A}$

Strain = $\frac{\Delta L}{L_0}$



Stress Strain graph



① OA $\rightarrow$ Stress $\propto$ Strain $\rightarrow$ Hooke's Law

$\frac{\text{Stress}}{\text{Strain}} = \text{modules of elasticity}$

② Beyond B $\rightarrow$ Plastic region, If Stress / Strain is beyond B point, it will loose elasticity.

③ F_{ext} removed, Permanent def will left.

④ All materials do not have Same graph

⑤ Breaking Strength of material

Hooke's Law ($\gamma, \beta, G = +ve \text{ value}$)

(N/m^2) $\frac{\text{Stress}}{\text{Strain}} = \text{modulus of Elasticity} [ML^{-1}T^{-2}]$

Young modulus

Bulk modulus

Shear modulus

$\gamma = \frac{\text{Longitudinal Stress}}{\text{Longitudinal Strain}}$

$\beta = \frac{\text{Volumetric Stress}}{\text{Volumetric Strain}}$

$G = \frac{\text{Shear Stress}}{\text{Shear Strain}}$

$\gamma = \frac{FL}{A\Delta L}$

$\beta = \frac{PV}{\Delta V}$

$G = \frac{F_T}{A\theta}$

③ Breaking Stress $\rightarrow$ ultimate Stress at which material breaks

$\hookrightarrow$ depends on material

$\hookrightarrow$ Independent of Length and Area

Breaking force

depends on length, Area + K

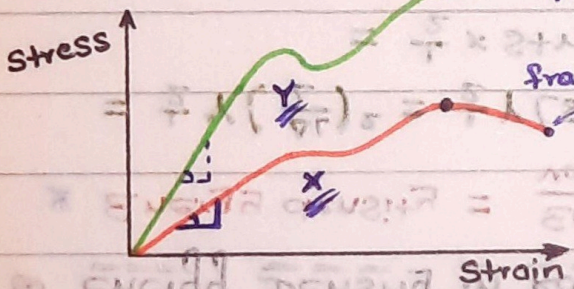
Compressibility

$K = \text{Compressibility} = \frac{1}{\beta}$

$\beta \downarrow K \uparrow$ (more compressible)

$\beta \uparrow K \downarrow$ (less compressible)

Stress / Strain



Fracture Point $\odot$ which has more strength & Breaking Stress $\odot$

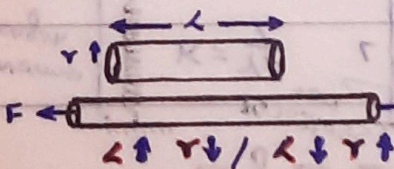
$\odot$ which is more malleable or brittle. $\odot$ more is necking region $\rightarrow$ more malleable

$\oplus$ more elastic will have more modulus of elasticity $\gamma \uparrow$

$\otimes \propto \text{slope} = \frac{\text{Stress}}{\text{Strain}}$

Slope $\uparrow \gamma \uparrow$ more elastic
Slope $\downarrow \gamma \downarrow$ less elastic

Poisson's Ratio



$\sigma = \frac{\text{Longitudinal Strain}}{\text{Lateral Strain}}$

$\sigma = \frac{\Delta R / R}{\Delta l / l}$

Relation b/w γ, K, η and σ

$\gamma = 3K(1 - 2\sigma), \gamma = 2\eta(1 + \sigma), \frac{\gamma}{\eta} = \frac{3}{\eta} + \frac{1}{K}$

$\frac{dp}{p} = \frac{dp}{\rho} \rightarrow \rho gh$

Relation b/w angle of Shear and Angle of Twist

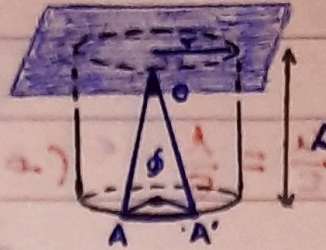
twist angle = ϕ Shear angle = θ

$$\phi = \frac{AA'}{Y}$$

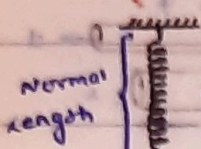
Arc Length = Angle
Radius

$$\theta = \frac{AA'}{L}$$

$$AA' = Y\phi = \theta L$$



Spring equivalent of Rod



Normal length

$$K = \frac{YA}{L}$$

massless rod

can be treated as spring

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F_r L}{A \Delta L}$$

$$F_r = \left(\frac{YA}{L}\right) \Delta L$$

$$K = \frac{YA}{L}$$



$$F_s = K \Delta L$$

$$F_r = K (\Delta L)$$

$$F_r = K (\Delta L)$$

$$F_r = K (\Delta L)$$

$$F_r = K (\Delta L)$$

$$F_r = K (\Delta L)$$

$$F_r = K (\Delta L)$$

$$F_r = K (\Delta L)$$

$$F_r = K (\Delta L)$$

Energy Density in Rod

$$\text{Energy Stored in spring} = \frac{1}{2} K x^2$$

$$\text{Energy density} = \frac{\text{Energy}}{\text{Volume}} = \frac{1}{2} \frac{YA}{L} \times \Delta L^2$$

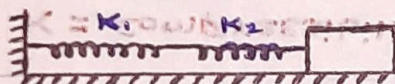
$$= \frac{1}{2} Y \left(\frac{\Delta L}{L}\right)^2 = \frac{1}{2} Y (\text{Strain})^2 = \frac{1}{2} \frac{(\text{Stress})^2}{Y}$$

$$= \frac{1}{2} \times \text{Stress} \times \text{Strain}$$

$$\frac{\text{Stress}}{\text{Strain}} = Y$$

$$\text{Energy} = (\text{Energy density}) \times \text{volume of Rod}$$

Series and Parallel of Springs

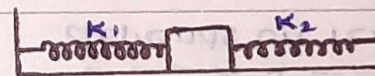


Series

$$\frac{1}{K_s} = \frac{1}{K_1} + \frac{1}{K_2}$$



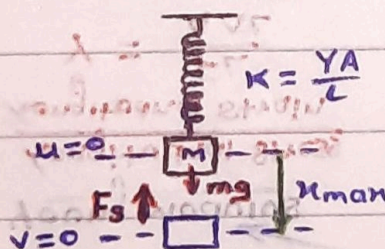
Parallel



$$K_p = K_1 + K_2$$

In case of rods $\rightarrow$ convert them to Spring $K = YA/L$

Release from Rest max Elongation



$$\text{directly } x_{\text{max}} = \frac{2mg}{K} = \frac{2mgL}{YA}$$

$$W_c + W_n = K_2 - K_1$$

$$+ mgh = \frac{1}{2} K x^2 = 0 - 0$$

$$x = \frac{2mg}{K}$$

For completely absorbing surface $P = \frac{v}{c}$

$$P = \frac{1}{2} \times \text{Energy flux} \times \text{Area} / c$$

Thermodynamics and KTG

* Thermometric Identity = $\frac{V(T) - V(\text{Ice})}{V(\text{Steam}) - V(\text{Ice})} \times 100^\circ\text{C}$

* $V_{\text{Ice}} = 20\text{cm}^3$

$V_{\text{Steam}} = 100\text{cm}^3$

when dip in $\Delta T \rightarrow$ Temp

we found $V = 60\text{cm}^3$

$$= \frac{60 - 20}{100 - 20} \times 100^\circ\text{C}$$

$$= \frac{40}{80} \times 100 = 50^\circ\text{C}$$

* Comparison b/w different Temperature scale

$$\frac{T(K) - 273}{373 - 273} = \frac{T^\circ\text{C} - 0}{100 - 0} = \frac{T^\circ\text{F} - 32}{212 - 32} = \text{const}$$

Q A thermometer with an arbitrary scale has the ice point at -20° and the steam point at 180° . when the thermometer reads 5° , a centigrade thermometer will read:

Ans $\frac{5 - (-20)}{180 - (-20)} = \frac{T^\circ\text{C} - 0}{100 - 0} \Rightarrow T^\circ\text{C} = 12.5^\circ\text{C}$

Thermal Expansion

* Linear expansion

$$\Delta L \propto \Delta T \propto L_0 \Rightarrow \Delta L \propto L_0 \Delta T$$

L_0 = Initial length

T = Initial Temp

$$\Delta L = \alpha L_0 \Delta T \Rightarrow L_f - L_0 = \alpha L_0 \Delta T$$

$$\alpha = \frac{\Delta L}{L_0 \Delta T} / ^\circ\text{C}$$

$$L_f = L_0 [1 + \alpha \Delta T]$$

$\alpha \rightarrow$ very small

* α = Linear coefficient of thermal expansion

* Superficial (areal) exp and Volumetric exp

$$L_f = L_0 (1 + \alpha \Delta T) \quad b_f = b_0 (1 + \alpha \Delta T) \quad h_f = h_0 (1 + \alpha \Delta T)$$

$$A_f = L_f b_f = A_0 [1 + 2\alpha \Delta T] \quad A_0 = l_0 b_0 \quad V_0 = l_0 b_0 h_0$$

$$A_f = A_0 [1 + \beta \Delta T] \quad \beta = \frac{\Delta A}{A \Delta T} \quad \beta = 2\alpha$$

$$V_f = V_0 [1 + \gamma \Delta T] \quad \gamma = 3\alpha$$

Effect of temp on the time period of a Simple pendulum

$$T_{\text{new}} = 2\pi \sqrt{\frac{L_f}{g}} = 2\pi \sqrt{\frac{L_0 [1 + \alpha \Delta \theta]}{g}} = T_0 [1 + \frac{\alpha \Delta \theta}{2}]$$

$$T_{\text{new}} = T_0 [1 + \frac{\alpha}{2} \Delta \theta] \quad T_0 = 2\pi \sqrt{L_0/g} \quad T \uparrow \theta \uparrow$$

$$\frac{T_{\text{new}} - T_0}{T_0} = \frac{\Delta T}{T_0} = \text{fractional change in Time} = \frac{\alpha}{2} \Delta \theta$$

$$\% \text{ change} = \frac{\alpha}{2} \Delta \theta \times 100 = 50 \alpha \Delta \theta$$

Imp T_0 (15) Heat T_1 (25)
 oscillate slowly Cool T_2 (0.5)
 fast oscillate

for a day = $24 \times 60 \times 60 \times \frac{\Delta T}{T}$

Time lost per sec = $\frac{\Delta T}{T_0} = \frac{\alpha}{2} \Delta \theta$

Time gain per sec = $\frac{\Delta T}{T_0} = \frac{\alpha}{2} \Delta \theta$

$\Delta T = \frac{1}{2} (\alpha \Delta T) T$

Effect on M.O.I

Any body in Quc Solid sphere $I = \frac{2}{5} MR^2$

$$I_f = \frac{2}{5} MR_f^2 = \frac{2}{5} MR^2 [1 + \alpha \Delta T]^2 = \frac{2}{5} MR^2 [1 + 2\alpha \Delta T]$$

$$I_f = I_0 [1 + 2\alpha \Delta T] = I_0 [1 + \beta \Delta T]$$

Shell of mass = M Radius = R Heat $T_{ext} = 0$
 $R_i = R_f$

$I_0 = \frac{2}{3} MR^2$ $I_0 \omega_0 = I_f \omega_f$

$I_0 \omega_0 = I_0 [1 + 2\alpha \Delta T] \omega_f$

$\omega_f = \frac{\omega_0}{[1 + 2\alpha \Delta T]} = \omega_0 [1 - 2\alpha \Delta T]$

Effect on Density

T_1 M volume = V_0 density = ρ_0 T $\uparrow$ volume = $V_f = V_0 [1 + \gamma \Delta T]$

with $\uparrow$ in temp $\rho \downarrow$ $\rho_f = \frac{M}{V_f} = \frac{M}{V_0 [1 + \gamma \Delta T]}$

$\rho_f = \rho_0 [1 - \gamma \Delta T]$

Effect on Buoyancy

whole system is heated to T_2 will be Rise, fall or remains at some place

density = ρ volume = V mass = $\rho V = \text{const}$ Float weight = F_B
 $V_b \rho_b g = V_b \rho_l g$

density σ $F_B = V_b \rho_l g = V_0 [1 + \gamma_b \Delta T] \frac{\sigma}{[1 + \gamma_l \Delta T]} g$

Temp = T_1 $F_B = \frac{V_0 \sigma g}{\rho_0} \frac{[1 + \gamma_b \Delta T]}{[1 + \gamma_l \Delta T]}$

$$\gamma_b > \gamma_l$$

$$F_B > B_0$$

(Rise)

$$\gamma_b = \gamma_l$$

$$F_B = B_0$$

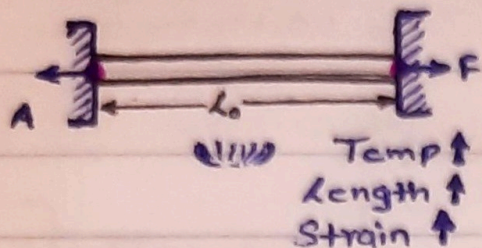
(Rest)

$$\gamma_b < \gamma_l$$

$$F_B < B_0$$

(Sink)

* Thermal Stress



(Thermal Stress)

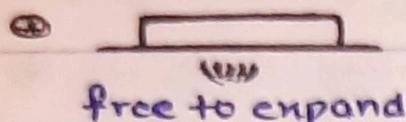
Young modulus = Y (Thermal Strain)

$$L_f = L_0 [1 + \alpha \Delta T] \Rightarrow \frac{\Delta L}{L_0} = \alpha \Delta T$$

$$Y = \frac{\text{Thermal Stress}}{\text{Thermal Strain}}$$

$$\text{Thermal Stress} = Y \frac{\Delta L}{L} = Y \alpha \Delta T$$

$$\text{Force} = \text{Thermal Stress} \times A = Y \alpha \Delta T A$$



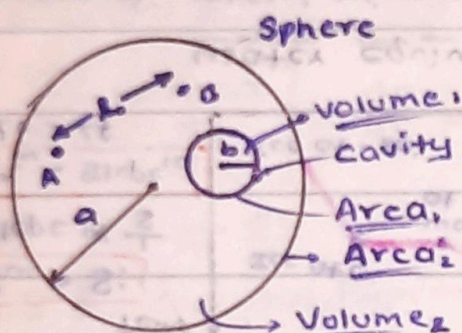
free to expand

Thermal Stress = ?

It is free to expand

there will be no stress

Expansion of Cavity



Final length b/w A and B
 $L_f = L_0 [1 + \alpha \Delta T]$

on heating

$$a_f = a [1 + \alpha \Delta T]$$

$$b_f = b [1 + \alpha \Delta T]$$

$$A_{1f} = A_1 [1 + \beta \Delta T]$$

$$A_{2f} = A_2 [1 + \beta \Delta T]$$

$$V_{1f} = V_1 [1 + \gamma \Delta T]$$

$$V_{2f} = V_2 [1 + \gamma \Delta T]$$

effect on heating

$(a \uparrow, b \uparrow) \rightarrow$ on heating outward flow

Expansion of fluids

Heat $\rightarrow$ container and liq both expand



$$V_{fc} = V_0 [1 + \gamma_c \Delta T] \quad V_{fl} = V_0 [1 + \gamma_l \Delta T]$$

$$\text{change in volume} = V_{fl} - V_{fc} = V_0 \Delta T [\gamma_l - \gamma_c]$$

Initial volume = V_0

$$\gamma_l > \gamma_c$$

overflow (+)

$$\gamma_l = \gamma_c$$

Remain same level 0

$$\gamma_l < \gamma_c$$

Level will fall (-)

$$\gamma_l - \gamma_c = \gamma_a$$

$$\gamma_a = \gamma_l + \gamma_c$$

Q A thin copper wire of length L increases in length by 1% when heated from temp T_1 to T_2 . what is the % change in area when a thin copper plate having dimensions $2L \times L$ is heated from T_1 to T_2 ?

$$\text{Ans } \frac{\Delta L}{L} \times 100 = 1 \quad \alpha = \frac{\Delta L}{L \Delta T} \quad A_f = A_0 [1 + \beta \Delta T]$$

$$\frac{\Delta L}{L} = 10^{-2}$$

$$\alpha = \frac{10^{-2}}{T_2 - T_1}$$

$$\frac{A_f - A_0}{A_0} = \beta \Delta T$$

$$\frac{\Delta A}{A} = \beta \Delta T$$

$$\% = 100 \beta \Delta T = 2\%$$

$$\beta = 2\alpha = \frac{2 \times 10^{-2}}{T_2 - T_1}$$

Q A hole is drilled in a Copper Sheet. $D_{\text{hole}} = 4.24 \text{ cm}$ at 27°C . What is the change in the diameter of the hole when the sheet is heated to 227°C ?
 Ans $d_f = d_o [1 + \alpha \Delta T] \Rightarrow d_f - d_o = d_o \alpha \Delta T$

Calorimetry

Specific heat (s or c) $\Rightarrow$ Heat for a body in specific conditions.

$$S = \frac{\Delta Q}{m \Delta T} \quad c_w = 1 \text{ cal/g}^\circ\text{C} = 4200 \text{ J/kg}^\circ\text{K}$$

Molar heat capacity

$$C = \frac{\Delta Q}{n \Delta T} \quad \text{J/mol}^\circ\text{K}$$

at const volume $\quad \quad \quad$ at const pressure

$$C_v = \frac{1}{n} \frac{dQ}{dT} \Big|_v$$

$$C_p = \frac{1}{n} \frac{dQ}{dT} \Big|_p$$

Thermal capacity

$$S = \frac{\Delta Q}{\Delta T}$$

$$mS = \frac{\Delta Q}{\Delta T} = \text{Heat capacity} \quad \text{cal/}^\circ\text{C} \text{ or } \text{cal/K}$$

Heat for Temp change $\quad \quad \quad$ Heat for State change

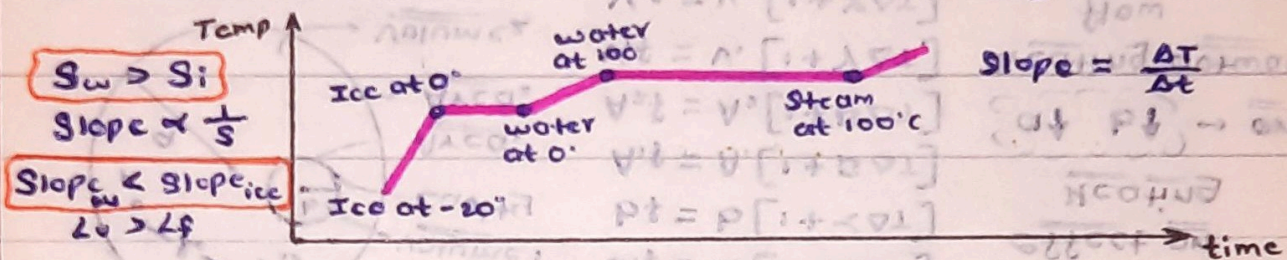
$$\Delta Q = mS \Delta T$$

$$\Delta Q = C \Delta T$$

$$C = mS$$

$$Q = mL$$

$$L = \text{Latent Heat} \quad \text{J/kg}$$



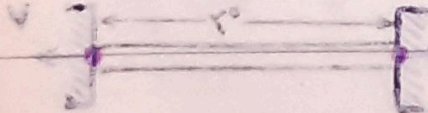
water equivalent of a body

* mass of water having the same thermal capacity as the body.

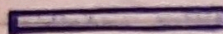


$$\text{Heat supply} = \text{Heat gained by } 300\text{g of } H_2O$$

$$= \frac{300}{1000} \times 4200 \times \Delta T$$



Heat Transfer

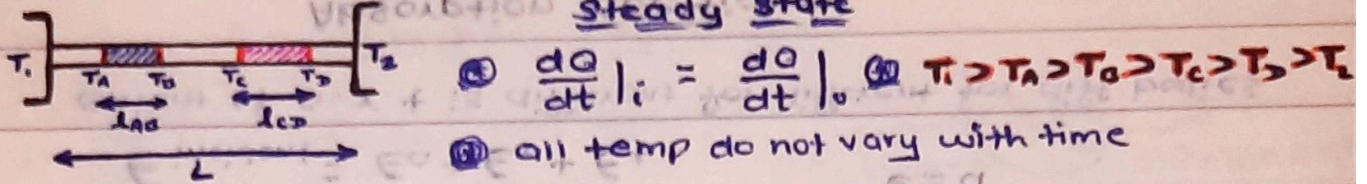
→ conduction


→ convection
 fluids

→ Radiation
 NO medium

Thermal conduction

Steady state



④ In Steady State ⑤ $\frac{T_A - T_B}{L_{AB}} = \frac{T_C - T_D}{L_{CD}} = \frac{T_1 - T_2}{L}$

Temp gradient b/w any two points for same conductor = Same

Temp Gradient = $\frac{\text{change of Temp}}{\text{Length}}$

* Steady State $\frac{dQ}{dt}$ = Rate of flow of heat

$\frac{dQ}{dt} \propto \text{Area}$
 $\frac{dQ}{dt} \propto \frac{dT}{dx}$

$\frac{dQ}{dt} = KA \frac{dT}{dx}$

K = coefficient of thermal conductivity
 unit $\rightarrow \frac{Jm}{Sm^2K} = \frac{J}{SmK}$

⑥ $S = \frac{1}{m} \frac{dQ}{dT}$

⑦ $S \downarrow$ low heat high temp Rise

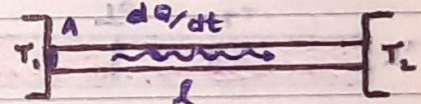
⑧ $S \uparrow$ High heat low temp change

$T_A = T_B$ If there is no flow of heat in AB

Analogy of conduction with ohm's law

* For Steady State conduction in Rod

$\frac{dQ}{dt} = KA \frac{dT}{dx} \neq T_1 - T_2$ analogy (voltage)



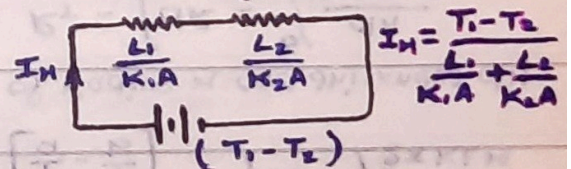
$\frac{dQ}{dt} = \frac{KA}{l} (T_1 - T_2)$ $\frac{dQ}{dt} = I_H$

$I_H = \frac{KA}{l} V_H \rightarrow V_H = I \left(\frac{l}{KA} \right)$

⑨ Equivalent conductivity for heat flow through slab in Serie

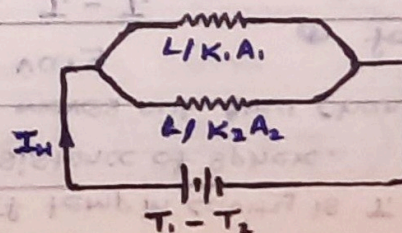
$R_{eq} = R_1 + R_2$

$\frac{L_1 + L_2}{KA} = \frac{L_1}{K_1 A} + \frac{L_2}{K_2 A}$ ⑩ $K_{eq} = ?$



⑪ $R_T = \frac{R_1 R_2}{R_1 + R_2} = \frac{\frac{L}{K_1 A_1} \cdot \frac{L}{K_2 A_2}}{\frac{L}{K_1 A_1} + \frac{L}{K_2 A_2}}$

$I_H = \frac{T_1 - T_2}{R_T}$



Q A sphere of thermal conductivity K . If temp in cavity is T_1 and in outer space is T_2 ($T_1 > T_2$). Find thermal resistance of sphere.

Ans

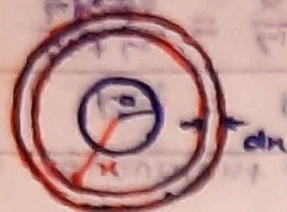


as heat current moves out Area change means $R \rightarrow$ vary

$$I_H = \frac{T_1 - T_2}{R_H} = \frac{T_1 - T_2}{\frac{1}{4\pi K} \left[\frac{1}{a} - \frac{1}{b} \right]}$$

for cylinder

$$\int dR = \int \frac{dn}{2\pi K L n}$$



we will take a shell of Radius n and thickness dn

$$dR = \frac{1}{K} \frac{dn}{4\pi n^2} \Rightarrow R_T = \int dR = \int_a^b \frac{dn}{4\pi K n^2} = \frac{1}{4\pi K} \left[-\frac{1}{n} \right]_a^b = \frac{1}{4\pi K} \left[\frac{1}{a} - \frac{1}{b} \right]$$

Q Consider two rods of equal cross sectional area A . one Al and Iron. Find the temp of junction of two rods in steady state.

Ans a) $R_H = R_1 + R_2$

$$= \frac{L_1}{K_1 A} + \frac{L_2}{K_2 A}$$

$$R_H = \frac{1}{A} \left[\frac{L_1}{K_1} + \frac{L_2}{K_2} \right]$$

$$b) I_H = \frac{T_1 - T_2}{\frac{1}{A} \left[\frac{L_1}{K_1} + \frac{L_2}{K_2} \right]}$$



$$I_1 = \frac{T_1 - T_0}{L_1 / K_1 A} \quad I_2 = \frac{T_0 - T_2}{L_2 / K_2 A}$$

$$I_1 = I_2$$

$$T_0 = \frac{\frac{K_1}{L_1} T_1 + \frac{K_2}{L_2} T_2}{\frac{K_1}{L_1} + \frac{K_2}{L_2}}$$

Radiation

$$T_0 = T_1 + (T_2 - T_1) \times \frac{n}{L}$$

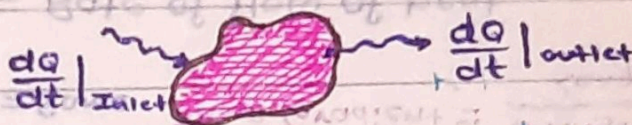
mode of heat transfer without any medium (EM radiations)

Prevost theory, when does a body cool or heat up

$$R_{inlet} > R_{outlet}$$

$$R_{inlet} = R_{outlet}$$

$$R_{inlet} < R_{outlet}$$



Q Krichoff's Radiation Law Good absorbers are Good emitters $e \approx a$

$$E_{incident} = E_a + E_r + E_t$$

extent of a, r, t is different for different for diff bodies

Absorption Coefficient

emissive coeff

absorptive coeff

$e =$ energy emitted by a body

energy absorbed by body

at temp T

$a =$

energy incident on body

energy emitted by a black body at same temp $= T$

$$0 < a < 1$$

Stefan's Law

Rate of flow of heat

$$\frac{dQ}{dt} = \sigma e A T^4$$

Temp of body (K)

Stefan's const
 5.67×10^{-8}

emissive
coefficient

Surface area of
body

⊗ Rate of cooling by Stefan's Law

① $\frac{dQ}{dt} \Big|_i = \frac{dQ}{dt} \Big|_o$ Temp const

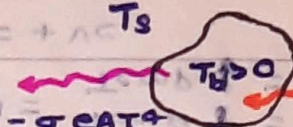
$$\sigma e A T_b^4 = \sigma e A T_s^4$$

$$T_b = T_s$$

$$e \approx A$$

$$\frac{dQ}{dt} = \sigma e A T_b^4$$

$$\frac{dQ}{dt} = \sigma e A T_s^4$$



② Rate of cooling $\frac{dQ}{dt} \Big|_o > \frac{dQ}{dt} \Big|_i = \frac{\lambda \times q \times b}{K L}$

$$\frac{dQ_{net}}{dt} = \frac{dQ}{dt} \Big|_o - \frac{dQ}{dt} \Big|_i \Rightarrow \frac{msdT}{dt} = \sigma e A (T_b^4 - T_s^4)$$

Rate of cooling $\frac{dT}{dt} = \frac{-\sigma e A}{ms} (T_b^4 - T_s^4)$

Newton's Law of cooling

we can use Approximation in Stefan's Law of cooling

$$\frac{dT}{dt} = -K (T_b - T_s)$$

$$T_b - T_s < 30K$$

different const

⊗ Consider a body having initial temp T_i in surro. of temp T_o .
Find the temp of body as a function of time and draw its graph.

Ans $\frac{dT}{dt} = -K (T - T_o)$

$$t = 0, T_i$$

$$t = t, T$$

Temp at any time

$$\int_{T_i}^T \frac{dT}{T - T_o} = \int_0^t -K dt$$

$$T = T_o + (T_i - T_o) e^{-Kt}$$

⊗ $t = 0, T = T_o + T_i - T_o = T_i$

⊗ $t = \infty, T = T_o$



Avg form of Newton's Law of cooling

Newton's Law of
cooling

$$\frac{dT}{dt} = -K (T - T_o)$$

change in temp

avg Temp of body

surrounding

we will use an avg form

$$\frac{T_2 - T_1}{t} = -K \left(\frac{T_1 + T_2}{2} - T_o \right)$$

$$t = 0, T_1$$

$$t = t, T_2$$

Wien's Displacement Law

① $\lambda_m \propto \frac{1}{T}$ — this is known as Wien's Displacement Law

$$\lambda_m T = b \quad \text{where } b \text{ is Wien const} = 2.89 \times 10^{-3} \text{ mK}$$

Relation b/w frequency and temp $\nu_m = \frac{c}{\lambda_m T}$

② A body changes its temp from 60°C to 50°C within 10 min by radiation. Find the time taken by body to change its temp from 40°C to 30°C . Const $T_s = 20^\circ\text{C}$

Ans $t = 0$ 10 min $t = ?$
 $t_1 = 60^\circ$ 50°C $40 \rightarrow 30$

$$\frac{50-60}{10} = -K \left(\frac{60+50}{2} - 20 \right) \quad \frac{30-40}{t} = -K \left(\frac{40+30}{2} - 20 \right)$$
$$\Rightarrow K = 1/35 \quad t = \text{--- min}$$

Maxwell's Law of Equipartition of Energy

for a molecule of gas each d.o.f contribute $\frac{1}{2}KT$ energy to total energy

① molecule has "f" d.o.f freedom

Ideal gas $P.E = 0$

② Energy of 1 molecule = $\frac{f}{2}KT$

$$T.I.E = K.E + P.E$$

③ Energy of 1 mole of molecule = $\frac{f}{2} (K \times N_A) T = \frac{f}{2} RT = \frac{f}{2} PV$

④ Energy of n moles of molecules = $\frac{n f R T}{2}$ (internal energy)

Internal Energy of an Ideal gas $\rightarrow$ function of Temp

$$du = \frac{n f R}{2} dt$$

$$P = \frac{F}{A} = \frac{2mv^2 n}{A}$$

Mean free Path

* Avg distance travelled by molecule b/w two successive colli.

$$\lambda_m = \frac{1}{\sqrt{2} \pi d^2 n}$$

$n = \frac{N}{V} = \text{No. of molecules per unit volume.}$

$$T_f = \frac{n_1 T_1 + n_2 T_2}{n_1 + n_2}$$

$d = \text{diameter of a molecule}$

Thermodynamics

$$\lambda = \frac{KT}{\sqrt{2} \pi d^2 p}$$

First Law $\rightarrow$ Energy Conservation $\Delta Q = \Delta U + \text{WORK}$

① $\Delta Q = \text{heat absorbed} = +ve$
heat Release = $-ve$

$\Delta U = \text{Temp } \uparrow = +ve$
Temp $\downarrow = -ve$

② work $\rightarrow$ Expansion ($V_f > V_i$) $w = +ve$
work done by Gas

$\rightarrow$ Compression ($V_f < V_i$) = $-ve$
work done on Gas

* Product of PV gives $\rightarrow$ Temp

Second Law of Thermodynamics

* Kelvin Planck Statement $\eta = 100\%$ System (not practically feasible)

* Claussius Statement - No process is possible whose sole result is the transfer of heat from a colder object to a hotter object.

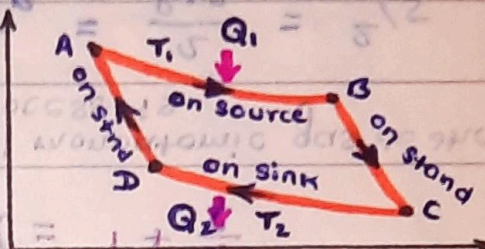
* Carnot Cycle - Carnot devised an ideal engine which is based on reversible cycle of four operations in succession.

1) Isothermal expansion $A \rightarrow B$

2) Adiabatic expansion $B \rightarrow C$

3) Isothermal compression $C \rightarrow D$

4) Adiabatic compression $D \rightarrow A$



Efficiency of a cycle (η)

$$\eta = \frac{\text{total mechanical work done by the gas in the whole process}}{\text{heat absorbed by the gas (only +ve)}} = \frac{\text{area of the P-V cycle}}{\text{heat injected into the system}}$$

* Heat engine - a device which converts heat into work.

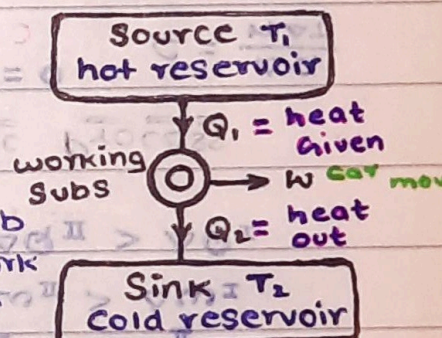
Three parts of a heat engine $\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$

(i) Source of high temp reservoir at temp T_1

(ii) Sink of low temp reservoir at temp T_2

(iii) Working substance $Q_1 = W + Q_2$

* In a cycle of heat engine the working substance extracts heat Q_1 from source, does some work W and rejects remaining heat Q_2 to the sink.



Efficiency of heat engine $\eta = \frac{\text{work done (W)}}{\text{heat taken from source (Q)}} \Rightarrow \frac{T_1 - T_2}{T_1} = \frac{Q_1 - Q_2}{Q_1}$

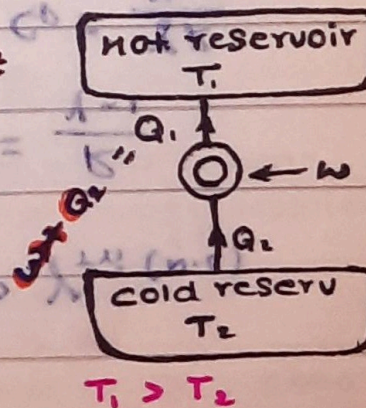
Refrigerator $\rightarrow$ It is the inverse of heat engine. It extracts (Q_2) from a cold res., some external work W is done on it and rejects heat (Q_1) to hot res. The coefficient of performance of a refrigerator (C.O.P. or β)

$$\beta = \frac{\text{Heat extracted from cold res}}{\text{work done on refrigerator}} \Rightarrow \frac{T_1 - T_2}{T_1} = \frac{Q_1 - Q_2}{Q_1}$$

For Carnot rever. refrigerator $\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$

$$\frac{dW}{dt} \Rightarrow \beta = \frac{T_2}{T_1 - T_2} = \frac{|Q_2|}{|Q_1| - |Q_2|} = \frac{1 - \eta}{\eta}$$

when Carnot engine is used in bridge, Coefficient of performance



Molar Specific heat

$$\frac{\Delta Q}{n \Delta T} = C$$

$$\gamma_{\text{mono}} > \gamma_{\text{di}} \approx \gamma_{\text{tri}} > \gamma_{\text{tri}} (n.g.)$$

$$(C_v)_{\text{mix}} = \frac{n_1 c_{v1} + n_2 c_{v2} + \dots}{n_1 + n_2 + \dots}$$

$$C_v = \frac{R}{\gamma - 1}$$

$$(C_p)_{\text{mix}} = \frac{n_1 c_{p1} + n_2 c_{p2} + \dots}{n_1 + n_2 + \dots}$$

$$C_p = \frac{\gamma R}{\gamma - 1}$$

$$\gamma_{\text{min}} = \frac{C_{p\text{min}}}{C_{v\text{min}}} = \frac{n_1 c_{p1} + n_2 c_{p2} + \dots}{n_1 c_{v1} + n_2 c_{v2} + \dots}$$

Molar Heat Capacities (C)

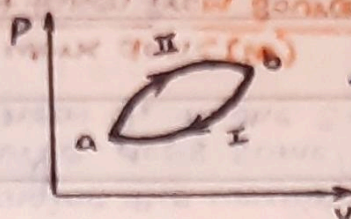
Ratio of
specific
heats

$$\frac{C_p}{C_v} = \gamma$$

$$\Delta Q = n C \Delta T$$

$$C = \frac{\Delta Q}{n \Delta T}$$

process
dependent



$$\Delta U_I = \Delta U_{II}$$

$$\Delta W_{II} > \Delta W_I$$

$$\Delta Q_{II} > \Delta Q_I$$

Isothermal Process

$$\Delta T = 0$$

$$C = \frac{\Delta Q}{n \Delta T}$$

$$C = \infty$$

Adiabatic Process

$$\Delta Q = 0$$

$$C = 0$$

$$C = \frac{\Delta Q}{n \Delta T}$$

Isochoric Process

$$V = \text{const}$$

$$\Delta W = 0$$

$$\Delta Q = n C_v \Delta T$$

$$\Delta Q = \Delta U$$

$$\Delta U = n C_v \Delta T$$

(valid in all process)

Isobaric Process

$$P = \text{const}$$

$$\Delta Q = n C_p \Delta T$$

$$\Delta W = P \Delta V = n R \Delta T$$

$$C_p = C_v + R$$

$$C_v = \frac{f}{2} R \quad \gamma = \frac{C_p}{C_v} \quad \gamma = 1 + \frac{2}{f}$$

the ratio of work done by an ideal monoatomic gas to the heat supplied to it in an isobaric process is

$$\text{Ans } \frac{\Delta W}{\Delta Q} = \frac{n R \Delta T}{n C_p \Delta T} = \frac{R}{C_p} = \frac{R}{\frac{f}{2} R + R} = \frac{2}{f+2} = \frac{2}{5}$$

Adiabatic Process

$$\Delta Q = 0$$

$$\beta = \gamma P$$

$$P V^\gamma = \text{const}$$

$$T V^{\gamma-1} = \text{const}$$

$$P^{1-\frac{1}{\gamma}} T = \text{const}$$

$$\frac{dP}{dV} = -\gamma \frac{P}{V}$$

slope

